

☒ Complete Simple Explanation of Cloud Kitchen P&L Intelligence Dashboard

☒ What Is This Project?

This project is a business analytics dashboard made for a company that operates many cloud kitchens across different cities.

The purpose of this project is to understand:

- How much money kitchens are earning
- How much profit they are making
- Which kitchens are performing well
- Which kitchens are wasting money
- Which cities are efficient
- Where operational problems exist
- How business decisions can be improved using data

This dashboard converts complicated Excel financial data into visual business insights that even non-technical people can understand easily.

☒ First Understand — What Is a Cloud Kitchen?

A cloud kitchen is a restaurant that only prepares food for online delivery.

It usually works with platforms like:

- Zomato
- Swiggy
- Uber Eats

These kitchens:

- Do not have dine-in customers
- Only prepare food for delivery
- Operate from centralized kitchens

A company may run hundreds of cloud kitchens across many cities.

Managing all of them manually becomes difficult.

That is why analytics dashboards become important.

☒ What Was The Data?

The dataset contained monthly financial and operational information of all kitchens.

Each row represented:

One kitchen + One month of performance

Example:

Month	Store	City	Revenue	EBITDA
Jan-2024	Kitchen A	Ahmedabad	₹40L	₹8L

This means:

Kitchen A in Ahmedabad generated:

- ₹40 Lakhs revenue
- ₹8 Lakhs operational profit

during January 2024.

☒ What Is P&L?

P&L means:

Profit and Loss

It is a financial statement that tells:

- How much money business earned
- How much money business spent
- Final profit or loss

It helps businesses understand:

- Are we making money?
 - Are operations efficient?
 - Which branches are profitable?
 - Where are we losing money?
-

📌 Understanding All Financial Terms In Simple Language

1 📌 NET REVENUE

What Is Net Revenue?

Net Revenue means:

Total money earned from selling food

It is the actual business generated by the kitchen.

Example

Suppose a kitchen sold:

- Pizza
- Burgers
- Rolls
- Drinks

Total monthly sales:

₹40 Lakhs

Then:

Net Revenue = ₹40 Lakhs

Why Is Revenue Important?

Revenue tells:

- How popular the kitchen is
- How much business it generates
- Customer demand
- Overall business growth

Higher revenue usually means:

- More customers
 - More orders
 - Strong market demand
-

2 📌 GROSS MARGIN (GM)

What Is Gross Margin?

Gross Margin tells:

How much money remains after direct food-related expenses

Direct costs include:

- Ingredients
- Vegetables
- Meat
- Packaging
- Cooking materials

Example

Suppose:

Revenue = ₹40 Lakhs

Food preparation cost:

₹16 Lakhs

Then:

Gross Margin = ₹24 Lakhs

Meaning:

After paying food preparation costs, ₹24 Lakhs remain.

Why Gross Margin Matters?

Gross Margin tells:

- Is food cost under control?
- Is pricing good?
- Is kitchen efficient?

If Gross Margin is low:

Possible problems:

- High raw material cost
- Food wastage
- Poor pricing
- Inefficient procurement

3× GROSS MARGIN %

What Is Gross Margin %?

This tells:

What percentage of revenue remains after direct costs

Formula:

$GM\% = \text{Gross Margin} / \text{Revenue} \times 100$

Example

Revenue = ₹40 Lakhs
Gross Margin = ₹24 Lakhs

Then:

$$\begin{aligned}\text{GM\%} &= 24 / 40 \times 100 \\ &= 60\%\end{aligned}$$

Meaning:

After direct operational costs, the kitchen keeps:

60% of its revenue

Why GM% Is Important?

It helps compare kitchens fairly.

Because:

A ₹1 crore kitchen and ₹20 lakh kitchen cannot be compared directly.

Percentage comparison makes analysis fair.

4× EBITDA

Most Important Financial Metric

What Is EBITDA?

EBITDA stands for:

Earnings Before Interest, Taxes, Depreciation and Amortization

But in simple language:

Actual operational profit of the kitchen

It tells:

After paying all operational expenses,
how much money is left?

What Expenses Are Included?

EBITDA considers costs like:

- Staff salary
- Rent
- Electricity
- Delivery operations
- Kitchen maintenance

- Packaging
- Daily operational expenses

Example

Suppose:

Revenue = ₹40 Lakhs

All operational expenses:

₹32 Lakhs

Then:

EBITDA = ₹8 Lakhs

Meaning:

After all operational expenses, kitchen earned ₹8 Lakhs profit.

Positive EBITDA

EBITDA > 0

Means:

Kitchen is profitable

Negative EBITDA

EBITDA < 0

Means:

Kitchen is losing money

Why EBITDA Is Very Important?

Investors and businesses use EBITDA because it shows:

- Actual business health
- Operational efficiency
- Real profitability

It removes accounting complexities.

5× EBITDA %

What Is EBITDA %?

This tells:

How efficiently revenue converts into profit

Formula:

$$\text{EBITDA\%} = \text{EBITDA} / \text{Revenue} \times 100$$

Example

Revenue = ₹40 Lakhs

EBITDA = ₹8 Lakhs

Then:

$$\begin{aligned}\text{EBITDA\%} &= 8 / 40 \times 100 \\ &= 20\%\end{aligned}$$

Meaning:

Kitchen converts:

20% of its revenue into actual profit

Why EBITDA % Is Powerful?

It helps compare kitchens regardless of size.

A smaller kitchen may have:

- Lower revenue
- But much better efficiency

6σ VARIANCE

Most Important Operational Metric

What Is Variance?

Variance means:

Difference between expected value and actual value

In cloud kitchens, variance usually means:

Mismatch between expected inventory/cost
and actual inventory/cost

Simple Example

Suppose:

Kitchen expected:

₹5,00,000 inventory cost

Actual inventory cost became:

₹5,30,000

Then:

Variance = ₹30,000

Why Does Variance Happen?

Variance may happen because of:

- Food wastage
- Inventory leakage
- Wrong stock management
- Theft/shrinkage
- Incorrect forecasting
- Over-ordering ingredients
- Operational inefficiency

Why High Variance Is Dangerous?

High variance means:

- Money leakage
- Poor operations
- Low efficiency
- Reduced profitability

Even if revenue is high, high variance can destroy profit.

7. VARIANCE %

What Is Variance %?

Variance percentage tells:

How large operational mismatch is relative to revenue

Formula:

$$\text{Variance\%} = \text{Variance} / \text{Revenue} \times 100$$

Example

Revenue = ₹40 Lakhs

Variance = ₹20,000

Then:

$$\text{Variance\%} = 20,000 / 40,00,000 \times 100$$

Very low percentage is good.

Higher percentage means operational inefficiency.

❏ What Was The Goal Of This Project?

The company wanted to answer:

Revenue Questions

- Which city generates highest revenue?
 - Which kitchens grow fastest?
 - Which stores perform poorly?
-

Profitability Questions

- Which kitchens are profitable?
 - Which kitchens are losing money?
 - Which city has best EBITDA?
-

Operational Questions

- Which kitchens waste most inventory?
 - Where operational leakages exist?
 - Which cities have high variance?
-

Strategic Questions

- Which kitchens should expand?
 - Which kitchens need restructuring?
 - Which locations should be optimized?
-

❏ What Did I Build?

I built:

An Interactive Business Intelligence Dashboard

using:

- Python
 - Pandas
 - Plotly
 - Streamlit
-

The Dashboard Contains 2 Main Systems

❏ 1❏ Kitchen Level P&L Dashboard

This section analyzes:

- Revenue
- Gross Margin
- EBITDA
- Store profitability

- City performance
- Operational efficiency

2 Variance Intelligence Dashboard

This section analyzes:

- Operational leakage
- Variance patterns
- High-risk kitchens
- Inventory mismatch
- Revenue buckets
- Variance buckets

Understanding Dashboard Insights

Revenue Trend Analysis

This chart shows:

- Monthly business performance
- Growth trend
- Seasonal demand

Insight Found

Revenue remained stable across months.

Meaning:

- Customer demand is consistent
- Business operations are stable

Gross Margin Trend

This measures operational efficiency.

Insight Found

Gross Margin stayed around:

58%

Meaning:

The business retains strong margin after direct food costs.

This indicates:

- Efficient procurement
- Controlled ingredient costs
- Healthy pricing strategy

EBITDA Trend

This measures actual operational profitability.

Insight Found

EBITDA remained around:



Meaning:

Business operations are financially healthy.

The kitchens are efficiently converting revenue into profit.

EBITDA Distribution

This chart divides kitchens into:

- Profitable kitchens
 - Loss-making kitchens
-

Insight Found

About:



of kitchens were profitable.

This is a very strong business indicator.

City-wise Performance

This compares:

- Revenue
- EBITDA%
- Profitability

across cities.

Insight Found

Ahmedabad performed best overall.

Mumbai had lower EBITDA%.

Possible reasons:

- Higher operational cost
 - Lower efficiency
 - Higher wastage
-

Store Performance Matrix

This scatterplot compares:

- Revenue
- EBITDA%
- Gross Margin%

Each bubble represents one kitchen.

Insight Found

Most kitchens followed healthy profitability patterns.

Some kitchens had:

- High revenue
- But low EBITDA

Meaning:

Revenue alone does not guarantee profitability.

Operational efficiency is equally important.

Top Performing Stores

These kitchens had:

- High revenue
- High EBITDA%
- Strong efficiency

These can become:

- Benchmark kitchens
 - Expansion models
-

Bottom Performing Stores

These kitchens showed:

- Weak profitability
- Low EBITDA%
- Possible operational problems

They may need:

- Cost reduction
 - Operational restructuring
 - Strategic review
-

Variance Bucket Analysis

Kitchens were grouped into categories:

Category	Meaning
< ₹15K	Low variance
₹15K–20K	Medium variance
₹20K–25K	High variance
> ₹25K	Very high variance

Insight Found

Many kitchens had high variance.

This suggests:

- Inventory inefficiency
- Food wastage

- Operational leakage
-

Variance vs Revenue Analysis

This compares:

- Revenue
 - Variance
 - EBITDA
-

Insight Found

Some high-revenue kitchens also had very high variance.

Meaning:

Even successful kitchens may waste large amounts of money operationally.

Heatmap Analysis

The heatmap shows:

Which cities have highest operational mismatch

Insight Found

Some cities consistently showed higher variance%.

This helps management:

- Focus audits
 - Improve operations
 - Reduce wastage
-

🔧 How I Built This Project

Step 1 — Data Cleaning

I cleaned:

- Missing values
 - Incorrect data types
 - Duplicate records
-

Step 2 — Feature Engineering

I created important business KPIs like:

- GM%
 - EBITDA%
 - Variance%
 - Revenue buckets
 - Variance buckets
-

Step 3 — Exploratory Data Analysis

I analyzed:

- Trends
- Profitability
- City comparisons
- Operational efficiency

Step 4 — Visualization

I created:

- Interactive charts
- Heatmaps
- Scatterplots
- KPI cards
- Distribution charts

Step 5 — Dashboard Development

I built the dashboard frontend using Streamlit.

Features include:

- Interactive filters
- Real-time KPI calculations
- Dynamic visualizations
- Multi-tab structure

☒ Why This Project Is Valuable

This project demonstrates:

- Business understanding
- Financial analytics
- Operational intelligence
- Data storytelling
- Dashboard development
- KPI engineering
- Strategic analytics

☒ Final Outcome

This dashboard transformed raw operational data into an executive-level business intelligence platform.

It helps businesses:

- Improve profitability
- Reduce wastage
- Optimize operations
- Monitor kitchen performance
- Make data-driven decisions

The project combines:

- Financial analysis
- Operational analytics
- Data visualization
- Business intelligence

into one complete solution.